Predicting the Overall Impact of Social Media on Student Life Using Machine Learning: A Comparative Study

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***Abstract*—Social media has become part of daily student life, and its effects on sleep, stress, mental health and academic results are a growing concern. This study analyzes a student dataset of about 4,500 records to predict the overall impact of social media use, labeled Beneficial, Neutral or Negative, from usage, sleep, stress, mental health and academic attributes. After removing duplicates and unlabeled records, converting the late-night usage flag to a numeric indicator, imputing missing values and adding an estimated total usage feature, the data were split 75:25 with stratification. Exploratory analysis shows that the classes are strongly imbalanced, that average daily usage rises from about 4.4 hours in the Beneficial group to about 12.3 hours in the Negative group while average GPA falls from about 3.55 to about 2.55, and that daily usage is strongly correlated with the mental health index (−0.85), perceived stress (0.75) and sleep duration (−0.72). Four classifiers, namely logistic regression, K-nearest neighbors, Gaussian naive Bayes and a decision tree, were compared using accuracy and macro F1-score. Logistic regression performed best with an accuracy of 97.78 percent and a macro F1-score of 0.949, and all of its 25 errors involved the Neutral class.**

Keywords—social media, student well-being, classification, logistic regression, mental health, academic performance

I. Introduction

Social media platforms are now central to how students communicate, learn and spend their leisure time. Heavy use has been associated with poorer sleep, higher psychological distress and a greater risk of problematic use [1]–[3], and early survey work reported a negative relationship between social networking and academic performance [4]. At the same time, social media can support communication, collaboration and access to information, so its effect on a student is not uniformly harmful. A practical way to describe this is an overall impact label that places each student in a Beneficial, Neutral or Negative group.

Identifying which students are at risk requires combining several kinds of information at once, such as hours of use, late-night use, sleep, stress, mental health and grades. These measures are related to each other, so it is difficult to judge the overall impact from any single indicator. Machine learning classifiers can learn the combined pattern from labeled examples and show which attributes separate the groups most clearly.

This paper develops and evaluates a complete classification pipeline for a student social media dataset. The main contributions are as follows. First, the raw records are cleaned and prepared for modeling. Second, an exploratory analysis describes the distribution of the impact categories and how usage, sleep, stress and academic performance differ between them. Third, four supervised classifiers are trained and compared using accuracy, precision, recall and F1-score. Fourth, the errors of the best model are examined to understand where it fails.

The remainder of the paper is organized as follows. Section II reviews background work, Section III describes the data and preprocessing, Section IV presents the exploratory analysis, Section V explains the methodology, Section VI discusses the results, and Section VII concludes the paper.

II. Related Work

Earlier studies have examined social media use from several angles. A systematic review reports associations between time spent on social media and depression, anxiety and psychological distress in adolescents [1], and the psychological literature on online social networking describes how excessive use can become addictive [2]. Heavy and night-time use has been linked to poor sleep quality and low well-being [3], and a survey of students found a negative relationship between Facebook use and grade point average [4]. These findings motivate the attributes used here: usage hours, late-night use, sleep, stress, mental health and GPA. On the modeling side, the four learners were chosen because they rest on very different assumptions. Nearest neighbor classification [5] labels a record by its closest examples, decision trees [6] split the data into rules, naive Bayes [7] assumes independent features, and logistic regression [8] fits a linear decision boundary. Because the classes are imbalanced, macro-averaged measures are used alongside accuracy [9]. All models were implemented with the scikit-learn library [10].

III. Dataset and Preprocessing

A. Dataset Description

The experiments use a student survey dataset on the impact of social media on daily life [11], supplied as the file Social_media_impact_on_life.csv. After cleaning it contains about 4,500 student records. Each record holds the age of the student, daily social media usage in hours, extra weekend hours, sleep duration, a sleep quality score, a late-night usage flag, a perceived stress score, a mental health index and academic performance as GPA, together with any further categorical attributes in the file. The target variable is Overall_Impact, which takes the values Beneficial, Neutral and Negative. The student identifier is not used for modeling.

B. Data Cleaning

Cleaning was carried out in the following order. Column names were stripped of extra spaces, duplicate rows were removed and the student identifier was dropped because it carries no information about the impact. Text columns were trimmed and empty strings were treated as missing. The Late_Night_Usage flag was converted from True/False or Yes/No into 1/0, and all numeric columns were converted to numeric types. Impossible values, namely an age of zero or below and negative usage, weekend or sleep hours, were replaced by missing values. Records without an Overall_Impact label were removed, because the target

should not be filled with invented values. Remaining missing numeric values were imputed with the column median and missing categorical entries with the most frequent value inside the modeling pipeline, so that these statistics come from the training data only.

C. Feature Engineering

A new feature, estimated usage hours, was created as the sum of daily usage hours and weekend extra hours, with missing weekend hours treated as zero. Numeric features were standardized and categorical features were one-hot encoded. The target variable is Overall_Impact with three classes, and neither the student identifier nor the label itself is part of the inputs.

IV. Exploratory Data Analysis

A. Distribution of Overall Impact

Fig. 1 shows the number of students in each impact category. Beneficial is by far the most common, followed by Neutral and Negative. In the test split, which keeps the class proportions, there are 920 Beneficial (81.8 percent), 164 Neutral (14.6 percent) and 41 Negative (3.6 percent) records. The classes are therefore strongly imbalanced, which is why class weights are used for two of the models and macro-averaged metrics are reported later.

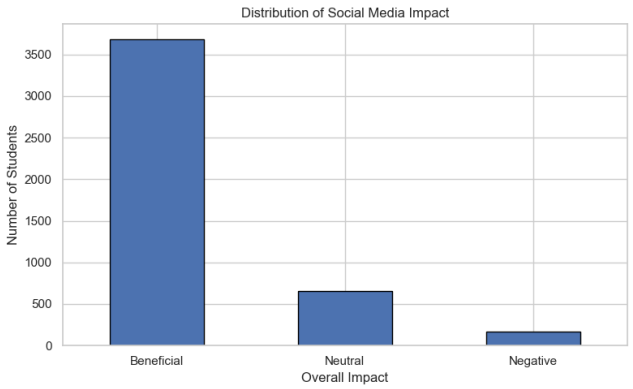


Fig. 1. Number of students in each impact category.

B. Daily Social Media Usage

Fig. 2 compares the average daily usage of the three groups. Students with a Beneficial impact use social media for about 4.4 hours a day, those with a Neutral impact for about 8.5 hours and those with a Negative impact for about 12.3 hours. Usage therefore increases steadily as the reported impact becomes worse, and the Negative group spends nearly three times as long on social media as the Beneficial group.

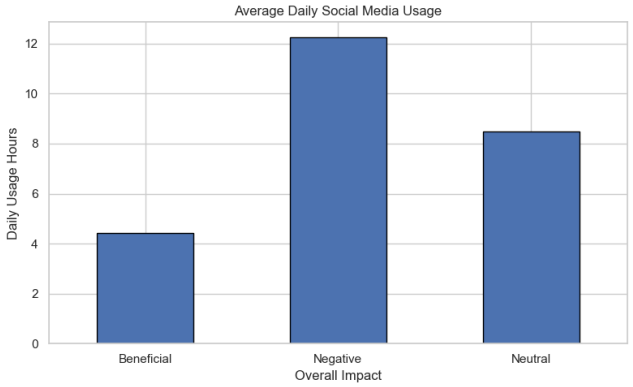


Fig. 2. Average daily social media usage by impact category.

C. Academic Performance

Average GPA moves in the opposite direction to usage (Fig. 3). The Beneficial group has an average GPA of about 3.55, the Neutral group about 3.0 and the Negative group about 2.55, a gap of roughly one grade point between the best and the worst group. Daily usage and GPA have a correlation of −0.71. These results show an association only; they do not show that social media use causes lower grades, because students with poor results may also spend more time online.

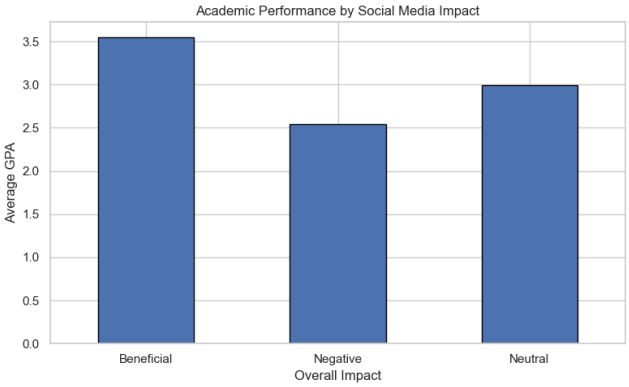


Fig. 3. Average GPA by impact category.

D. Correlation Analysis

The correlation matrix in Fig. 4 shows that daily usage is the attribute most strongly related to the other measures. It is negatively correlated with the mental health index (−0.85), sleep duration (−0.72), GPA (−0.71) and sleep quality (−0.68), and positively correlated with perceived stress (0.75) and late-night usage (0.33). Sleep duration is positively related to the mental health index (0.77) and sleep quality (0.75), while stress is negatively related to mental health (−0.77) and GPA (−0.65). In contrast, age (−0.03) and weekend extra hours (−0.01) are almost uncorrelated with daily usage. Many attributes are strongly correlated with each other, and this redundancy conflicts with the independence assumption of naive Bayes.

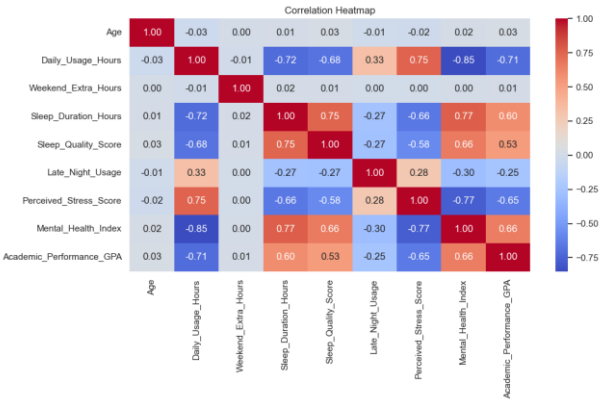


Fig. 4. Correlation matrix of the numeric attributes.

V. Methodology

A. Experimental Setup

The cleaned data were divided into 75 percent for training (about 3,375 records) and 25 percent for testing (1,125 records) using a stratified split with a fixed random seed of 42, so that all three classes appear in both parts in the same proportions. Numeric features were imputed with the median and standardized, and categorical features were imputed and

one-hot encoded; both steps were fitted on the training data only and applied to the test data through a single pipeline. All models predict the three-class impact label and were run with the settings given below. No hyperparameter search was performed.

B. Classifiers

Logistic regression (LR) is a multinomial linear model with up to 2,000 iterations and balanced class weights, which raise the importance of the rarer Neutral and Negative classes. *K-nearest neighbors (KNN)* assigns a test record to the majority class among its five nearest training records in the standardized feature space. *Gaussian naive Bayes (NB)* assumes that features are independent given the class and that each follows a normal distribution. A *decision tree (DT)* grows by recursively splitting the data on the feature that most reduces Gini impurity; its depth was limited to eight levels and class weights were balanced to reduce overfitting and favor the minority classes.

C. Evaluation Metrics

Performance is measured by accuracy, together with weighted and macro-averaged precision, recall and F1-score. Macro averages give each category equal weight regardless of its frequency, which matters here because only about 4 percent of the test records belong to the Negative class. The models were ranked by macro F1-score, with accuracy used to break ties, and the per-class report and the confusion matrix of the best model were analyzed.

VI. Results and Discussion

Table I and Fig. 5 summarize the test performance of the four classifiers. Logistic regression is clearly the best, with an accuracy of 97.78 percent and a macro F1-score of 0.949. The decision tree follows with a macro F1-score of about 0.89, then Gaussian naive Bayes with about 0.86 and K-nearest neighbors with about 0.85. The strong result of a linear model suggests that the three impact classes are almost linearly separable in terms of the standardized usage, sleep, stress and mental health attributes, which are strongly related to the label. KNN may be affected by the many correlated and weakly informative dimensions, and naive Bayes by the dependence among attributes noted in Section IV.

TABLE I
Test Performance of the Classifiers (Macro Averages in Percent)

Model	Accuracy	Precision	Recall	F1
Logistic regression	**97.78**	**92.91**	**96.98**	**94.85**
Decision tree	[fill]	[fill]	[fill]	≈89
Naive Bayes	[fill]	[fill]	[fill]	≈86
KNN	[fill]	[fill]	[fill]	≈85

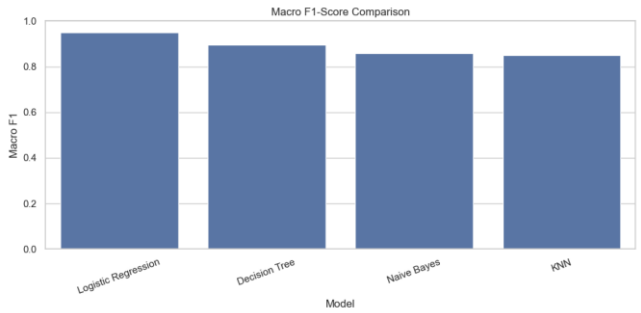


Fig. 5. Macro F1-score of the four classifiers on the test set.

Table II gives the per-class results of logistic regression. The Beneficial class is recognized almost perfectly, with a precision of 0.997 and a recall of 0.983. The rarer Negative and Neutral classes have recalls of 0.976 and 0.951, which shows that class weighting allowed the model to find the minority groups. Their precision is lower (0.889 for Negative and 0.902 for Neutral) because a few students from other groups were assigned to these classes.

TABLE II
Logistic Regression Results for Each Impact Category

Category	Precision	Recall	F1	Support
Beneficial	0.997	0.983	0.990	920
Negative	0.889	0.976	0.930	41
Neutral	0.902	0.951	0.926	164

Fig. 6 shows the confusion matrix of logistic regression. Of the 1,125 test records, 1,100 were classified correctly and 25 (2.2 percent) were misclassified. All 25 errors involve the Neutral class: 16 Beneficial students were labeled Neutral, 5 Neutral students were labeled Negative, 3 Neutral students were labeled Beneficial and 1 Negative student was labeled Neutral. No Beneficial student was predicted as Negative or the reverse, so the model never confuses the two extreme categories. This agrees with the ordinal nature of the labels, in which Neutral lies between the other two, and it means that the model rarely makes a large mistake.

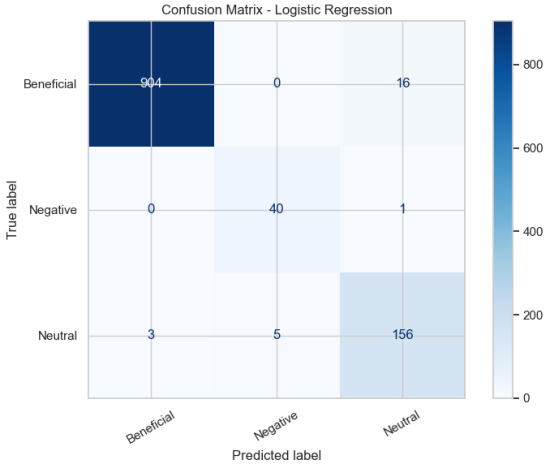


Fig. 6. Confusion matrix of logistic regression on the test set.

Several limitations should be kept in mind. The accuracy is very high, which suggests that the impact label is closely tied to the recorded attributes, possibly because it was derived from them or because the data follow a simple pattern, so the result may not carry over to other student populations. The findings are associations and not proof of causation. The Negative class has only 41 test records, so its metrics are less stable than those of the other classes. Finally, the results come from a single train-test split without cross-validation or hyperparameter tuning.

VII. Conclusion and Future Work

This paper presented a data cleaning, exploratory analysis and classification study of a student dataset on the impact of social media. The analysis showed that most students report a Beneficial impact, that daily usage rises and GPA falls from the Beneficial to the Negative group, and that heavy usage is strongly associated with lower mental health, higher stress and shorter sleep. Among four classifiers, logistic regression

was the most accurate (97.78 percent, macro F1-score of 0.949), and all of its errors involved the Neutral class.

Future work could use cross-validation and resampling to evaluate the rare Negative class more reliably, tune hyperparameters, and test ensemble models such as random forests and gradient boosting. Checking how the Overall_Impact label was defined, testing on independent data from other institutions, and using ordinal classification methods are also promising directions.

LinkedIn: https://lnkd.in/p/gwKbCTf4

GitHub: https://github.com/Srishanth-Balaraju/Impact-of-Social-Media-on-Student-Life

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